Nithyasri Rajulu

M.Sc. Mechatronics, KTH | Robotics, autonomous systems and computer vision

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Profile

Mechatronics master's student at KTH with a B.E. in Robotics and Automation and industrial experience in product design, drives and control. Work spans ROS 2 simulation, mechanical design, Python/OpenCV applications and robot integration. Interested in AI and reinforcement learning for autonomous systems. Seeking a 30 ECTS master's thesis in Sweden, Spring 2027.

Professional experience

Teaching Assistant, C Programming | KTH Royal Institute of Technology

August 2026-present | Stockholm, Sweden

- Support students with C assignments, debugging, pointers, memory management and embedded programming concepts.

Product Design and Development Intern | Prime Conveyors

May-July 2026 | Coimbatore, India

- Mechanical design for vertical carousel storage and AGV pallet concepts, including CAD and manufacturing drawings.
- Applied DFMA, component standardisation and assembly-sequence planning across product variants.

Drives and Automation Intern | SEW-Eurodrive

January-May 2025 | Sriperumbudur, India | Bachelor's thesis placement

- Built a MATLAB/Python GUI for radar servo-drive simulation; calculated drive parameters, compared responses and tuned P/PI controllers.
- Drive configuration using MOVISUITE and operator-interface development using MOVITOOLS.

Engineering Intern | Visual Education Aids

June-November 2024 | Coimbatore, India

- Supported motor and drive selection, mechanical integration and remote control for a motorised manipulator intended for hazardous-material handling.

Design Intern | Prime Precisions

May-June 2024 | Coimbatore, India

- Contributed to conveyor redesigns for customer-specific material-handling applications.

Education

M.Sc. Mechatronics | KTH Royal Institute of Technology

August 2025-14 June 2027 (expected graduation) | Stockholm, Sweden

- Coursework includes robotics, dynamics and motion control, embedded systems, robust mechatronics, and a mechatronic capstone project.

B.E. Robotics and Automation | PSG College of Technology

August 2021-May 2025 | Coimbatore, India

- Bachelor's thesis: Control System Simulation for Radar and Anti-sway Testing in EOT Cranes, completed at SEW-Eurodrive.

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Selected projects and technical skills

View my engineering portfolio online

Selected engineering projects

Industrial Computer Vision Demonstrators

October 2026 | Python, OpenCV

- Projects covering conveyor part counting, reference-based surface inspection and perspective-calibrated planar measurement.

Gesture-Controlled Virtual Calculator

Academic project reconstructed October 2026 | Python, OpenCV, MediaPipe

- Calculator interface with fingertip targeting, pinch input, arithmetic parsing and a mouse fallback.

Worm-Drive Gripper

2024 | Mechanical design team | SolidWorks

- Worm-drive mechanism, actuation concept and SolidWorks modelling for a robotic gripper.

Warehouse Mobile Manipulation

ROS 2 course project | 2026 | Nav2, py_trees, numerical IK

- Course implementation combines mobile-base navigation, a six-joint arm and vacuum pick-and-place, with sequential and behaviour-tree mission modes.
- Project focus includes coordinate transforms, numerical inverse kinematics and navigation/manipulation interfaces.

Technical skills

Robotics and simulation: ROS 2, Nav2, Gazebo, mobile robot navigation, robot integration, scenario simulation.

Programming and computer vision: Python, C, Embedded C, OpenCV, Bash, Linux, GUI development.

Control and automation: MATLAB, Simulink, P/PI tuning, MOVISUITE, MOVITOOLS, Siemens and Delta PLC, ladder logic, SCADA.

Mechanical design: SolidWorks, Fusion 360, AutoCAD, DFMA, manufacturing drawings, 3D scanning, FDM printing, hydraulics and pneumatics.

Certifications and leadership

Festo Industrial Pneumatic Control P111 (2024); SolidWorks intensive course (2020); AutoCAD (2020). IEEE student executive (2022-2025).

Languages

English, Tamil, Telugu and Hindi; German A1.